

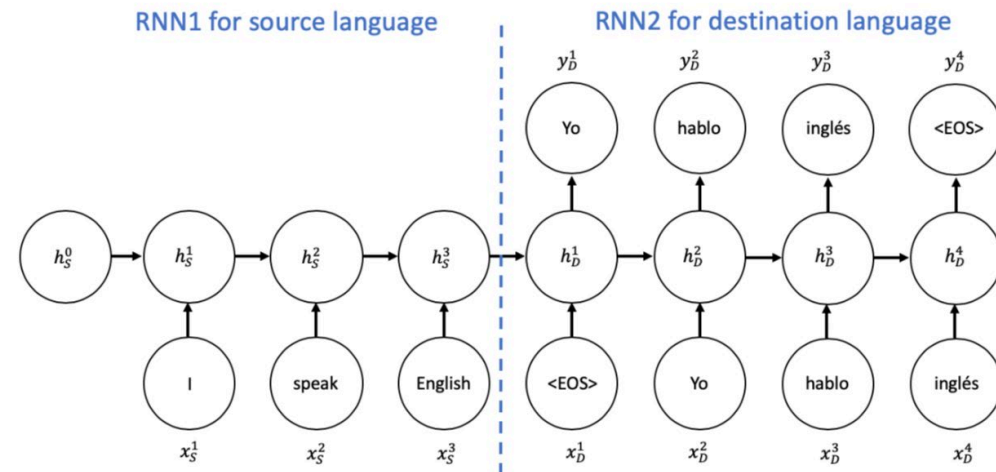
Attention Mechanism

Traditional RNN:

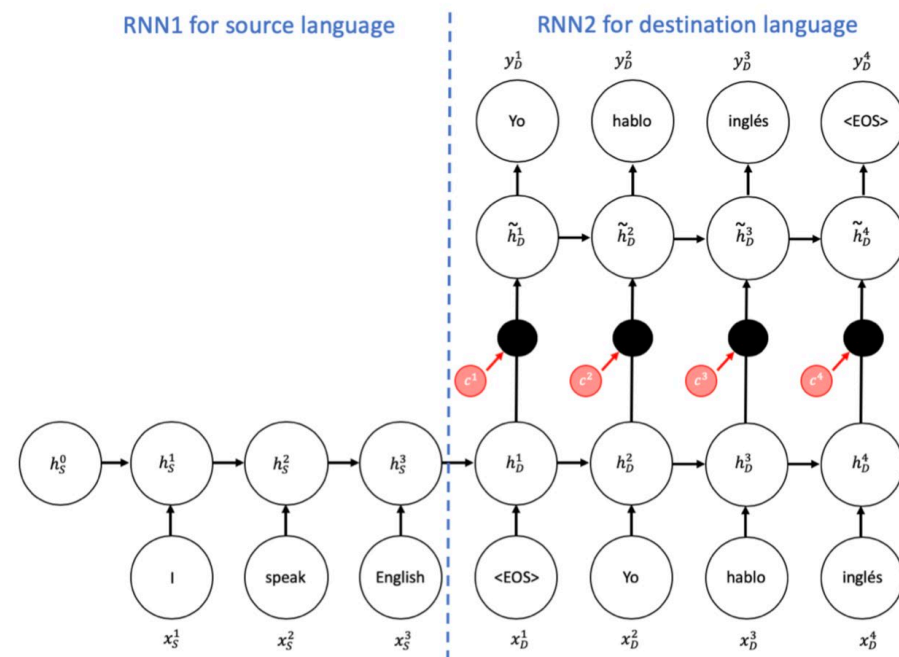
- the entire source sentence $\rightarrow$ a summary vector
- Influence of summary vector decreases as 2nd RNN progresses

Key Idea of Attention Mechanism: context vectors

- Augment hidden state of 2nd RNN w/ context vectors
- Context vector c^j (for position j) attends to most relevant hidden states of 1st RNN (pink node)



- Machine translation by tradition RNN



- Machine translation by attention+ RNN

Attention: Context Vectors

□ Ideal Case: perfect alignment

- If j^{th} word in source $\leftrightarrow j^{\text{th}}$ target word

$$c^j = h_S^j$$

□ In Reality

- Introduce an alignment vector a

$$a(i) = \text{Rel}(h_S^i, h_D^j) \quad \sum_i a(i) = 1 \quad \text{and} \quad a(i) \geq 0$$

- Linear weighted sum to obtain context vector

$$c^j = \sum_i a(i) h_S^i \quad a(i) = \frac{\exp(\text{score}(h_S^i, h_D^j))}{\sum_{i'} \exp(\text{score}(h_S^{i'}, h_D^j))}$$

- Score(): un-normalized relevance score (e.g., dot product, bi-linear, MLP)

□ Applications in NLP: Transformer, BERT

